



# Cambridge International AS & A Level

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**PHYSICS**

**9702/34**

Paper 3 Advanced Practical Skills 2

**October/November 2020**

**MARK SCHEME**

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

**PUBLISHED****Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

**GENERIC MARKING PRINCIPLE 1:**

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

**GENERIC MARKING PRINCIPLE 2:**

Marks awarded are always **whole marks** (not half marks, or other fractions).

**GENERIC MARKING PRINCIPLE 3:**

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

**GENERIC MARKING PRINCIPLE 4:**

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

**GENERIC MARKING PRINCIPLE 6:**

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

**Science-Specific Marking Principles**

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2 | The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3 | Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4 | The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5 | <p><u>'List rule' guidance</u></p> <p>For questions that require <b><i>n</i></b> responses (e.g. State <b>two</b> reasons ...):</p> <ul style="list-style-type: none"> <li>• The response should be read as continuous prose, even when numbered answer spaces are provided.</li> <li>• Any response marked <i>ignore</i> in the mark scheme should not count towards <b><i>n</i></b>.</li> <li>• Incorrect responses should not be awarded credit but will still count towards <b><i>n</i></b>.</li> <li>• Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should <b>not</b> be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.</li> <li>• Non-contradictory responses after the first <b><i>n</i></b> responses may be ignored even if they include incorrect science.</li> </ul> |

**6** Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g.  $a \times 10^n$ ) in which the convention of restricting the value of the coefficient ( $a$ ) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

**7** Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

| Question | Answer                                                                                                                                                                                                                                                                 | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(a)     | Value of raw $x$ to the nearest mm <b>and</b> final value in range 38.0–42.0 cm.                                                                                                                                                                                       | 1     |
| 1(b)     | Value of $\theta$ in range $60^\circ$ – $100^\circ$ .                                                                                                                                                                                                                  | 1     |
| 1(c)     | Six (or more) sets of readings of $x$ and $\theta$ with correct trend and without help from the Supervisor scores 5 marks, five sets scores 4 marks etc.<br><br>Values of $x$ should be different and non-zero.                                                        | 5     |
|          | Range: $x_{\min} \leq 20.0$ cm <b>and</b> $x_{\max} \geq 45.0$ cm.                                                                                                                                                                                                     | 1     |
|          | Column headings:<br>Each column heading must contain a quantity, a separating mark and a unit where appropriate.<br>Heading for $\cos\theta$ must have no unit.<br>The presentation of quantity and unit must conform to accepted scientific convention e.g. $x$ / cm. | 1     |
|          | Consistency:<br>All values of raw $\theta$ must be given to the nearest degree.                                                                                                                                                                                        | 1     |
|          | Significant figures:<br>Values of $\cos\theta$ should be to the same number of s.f. as (or one more than) the number of s.f. in the corresponding value of $\theta$ .                                                                                                  | 1     |
|          | Calculation:<br>Values of $\cos\theta$ calculated correctly.                                                                                                                                                                                                           | 1     |

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| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                     | Marks    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(d)(i)   | <p><b>Axes:</b><br/>Sensible scales must be used, no awkward scales (e.g. 3:10 or fractions).<br/>Scales must be chosen so that the plotted points occupy at least half the graph grid in both x and y directions.<br/>Scales must be labelled with the quantity that is being plotted.<br/>Scale markings should be no more than three large squares apart.</p>                           | <b>1</b> |
|           | <p><b>Plotting of points:</b><br/>All observations must be plotted on the grid.<br/>Diameter of plotted points must be <math>\leq</math> half a small square.<br/>Plots must be accurate to within half a small square in both x and y directions.</p>                                                                                                                                     | <b>1</b> |
|           | <p><b>Quality:</b><br/>All points in the table must be plotted (at least 5) for this mark to be awarded.<br/>Trend of points should be correct.<br/>Scatter of plotted points must be no more than <math>\pm 0.05</math> from a straight line on the <math>\cos \theta</math> axis.</p>                                                                                                    | <b>1</b> |
| 1(d)(ii)  | <p><b>Line of best fit:</b><br/>Judged by balance of all points on the grid (at least 5) about the candidate's line. There must be an even distribution of points either side of the line along the full length.<br/>One anomalous point is allowed only if clearly indicated (i.e. circled or labelled) by the candidate.<br/>Lines must not be kinked or thicker than half a square.</p> | <b>1</b> |
| 1(d)(iii) | <p><b>Gradient:</b><br/>The hypotenuse of the triangle used must be greater than half the length of the drawn line.<br/>Method of calculation must be correct, e.g. not <math>\Delta x / \Delta y</math>.<br/>The sign of the gradient must match the graph drawn.<br/>Both read-offs must be accurate to half a small square in both the x and y directions.</p>                          | <b>1</b> |
|           | <p><b>y-intercept:</b><br/>Correct read-off from a point on the line substituted into <math>y = mx + c</math> or an equivalent expression.<br/>Read-off accurate to half a small square in both x and y directions.<br/><b>or</b><br/>Intercept read directly from the graph, with read-off at <math>x = \text{zero}</math> accurate to half a small square in y direction.</p>            | <b>1</b> |

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| Question | Answer                                                                                                                    | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------|-------|
| 1(e)     | $a$ equal to candidate's gradient <b>and</b> $b$ equal to candidate's intercept. Values must not be written as fractions. | 1     |
|          | Unit for $a$ correct (e.g. $\text{cm}^{-1}$ ) and consistent with value <b>and</b> no unit given for $b$ .                | 1     |

| Question  | Answer                                                                                                                                                                                                                                                                                            | Marks |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(a)(i)   | Value of raw $x$ to nearest mm <b>and</b> final value in range 20–24 mm with unit.                                                                                                                                                                                                                | 1     |
| 2(a)(ii)  | Values of raw $D$ and $d$ to nearest mm <b>and</b> final value of $D$ in the range 38–42 mm.                                                                                                                                                                                                      | 1     |
| 2(a)(iii) | Correct calculation of $L$ .                                                                                                                                                                                                                                                                      | 1     |
| 2(a)(iv)  | Justification based on significant figures in $x$ , $D$ and $(D - d)$ .                                                                                                                                                                                                                           | 1     |
| 2(b)(i)   | Raw values of times all to 0.1 s or all to 0.01 s <b>and</b> value of $T$ in range 0.5–2.5 s.                                                                                                                                                                                                     | 1     |
|           | Evidence of repeats: at least two values of time.                                                                                                                                                                                                                                                 | 1     |
| 2(b)(ii)  | Percentage uncertainty based on an absolute uncertainty in time value of 0.2–0.5 s.<br>If repeat readings have been taken, then the absolute uncertainty can be half the range (but not zero) if the working is clearly shown.<br>Correct method of calculation to obtain percentage uncertainty. | 1     |
| 2(c)      | Second values of $x$ and $T$ .                                                                                                                                                                                                                                                                    | 1     |
|           | Quality: second $T >$ first $T$ .                                                                                                                                                                                                                                                                 | 1     |
| 2(d)(i)   | Two values of $k$ calculated correctly. The final values of $k$ must not be fractions.                                                                                                                                                                                                            | 1     |
| 2(d)(ii)  | Valid comment relating to the calculated values of $k$ , testing against a criterion specified by the candidate.                                                                                                                                                                                  | 1     |
| 2(e)      | Correct calculation of $g$ .                                                                                                                                                                                                                                                                      | 1     |



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| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(f)(i)  | <p>A Two readings are not enough to draw a (valid) conclusion (<b>not</b> “not enough for accurate results”, “few readings”).</p> <p>B Difficult to measure diameters or <math>x</math> with reason e.g. parallax/zero not at end of ruler/<math>x</math> varies around roller/nuts or bolt prevent placement of ruler.</p> <p>C <math>\theta</math> small so large uncertainty/large % uncertainty in <math>\theta</math>.</p> <p>D Oscillations die away quickly.</p> <p>E Difficult to measure time/<math>T</math> because it is difficult to judge the end/beginning/completion of an oscillation.</p> <p>F Roller slips down board.</p> <p><i>1 mark for each point up to a maximum of 4.</i></p>                                                                                                                           | <b>4</b> |
| 2(f)(ii) | <p>A Take more readings <u>and</u> plot a graph or take more readings <u>and</u> compare <math>k</math> values (<b>not</b> “repeat readings” on its own).</p> <p>B Use metre rule/calipers/micrometer/blocks either side of washer/use pointers with detail.</p> <p>C Method to reduce uncertainty in <math>\theta</math> e.g. use of trigonometry with detail/increase <math>\theta</math>.</p> <p>D Method to increase the number of oscillations e.g. smoother board/named smooth material for board/sand board.</p> <p>E Method of improving timing e.g. fiducial marker at centre of oscillation or video/record/film with timer/frame-by-frame.</p> <p>F Method of reducing slipping e.g. rougher/named higher friction surface/smaller <math>\theta</math>.</p> <p><i>1 mark for each point up to a maximum of 4.</i></p> | <b>4</b> |